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FORA Communication Protocol for WS series

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Revision History.

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3 2012/05/07 Carol Modification. Ver1.02
4 2012/07/23 Carol Command 71, 72 code / user ID value Ver1.03
5 2012/07/24 Carol Modification. Ver1.04
6 2012/09/11 Carol Command 72. Ver1.05
7 2013/03/27 Carol Byte 4 in 0x71 command Ver1.06
8 2013/06/03 Carol Command 0x54 modification. Ver1.07
9 2015/01/19 Carol Command 0x71 modification. Ver1.08
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1. General.

1.1. Overview .

The document describes the commands which will be used to communicate with 25xx series.

Before retrieving data in the device, please make sure you have already established the proper connection channel between the device and the host side. For example, if you use USB cable, be sure that the USB driver has been properly installed; or if you use Bluetooth communication, be sure that the connection has been successfully established.

Generally speaking, the device will be the slave side waiting for the request from the host which is usually the computer or the hub/gateway.

Note for different Bluetooth versions:

For Bluetooth v2 meter: Please connect to meter by SPP (Serial Port Profile). Meter always stands by for both pairing and connection, the PIN for pairing is 111111.

For Bluetooth v3 meter: Please connect to meter by SPP (Serial Port Profile). Meter must be paired before any connection, please find the instruction in meter manual for the pairing process. The meter supports SSP (Simple Secure Pairing), as a result PIN is not required. If host does not support SSP, please use 0000 as PIN if required.

For Bluetooth v4 meter: Please connect to meter by following GATT service:

UUID Base: 1212-efde-1523-785feabcd123

Service UUID: 0x1523

Characteristic: 0x1524 (write/notify)

1.2. Document Overview.

For the first time user on the device integration, please read the 2.1 frame structure. If you are familiar with serial port communication, you can review the command list table in section 2.2 directly. The following sub sections are all about the details for each command.

1.3. Definitions, Acronyms, and Abbreviations.

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GW Gateway / Hub / Computer side

MD Medica Device, ie. blood glucose meter, blood pressure meter or scales,
etc.

CMD Command

NA Not available

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2. Interface specification

Block Diagram

The communication interface is base on RS232 serial port communication :

2.1. Frame Structure

8-byte frame

Byte 1 2 3 4 5 6 7 8

Name Start CMD Data_0 Data_1 Data_2 Data_3 Stop ChkSum

Format HEX

GW MD 51 CMD Data_0 Data_1 Data_2 Data_3 A3 [1..7]

GW MD 51 ACK Data_0 Data_1 Data_2 Data_3 A5 [1..7]

Variable frame

Byte 1 2 3 4 ~ n-2 n-1 n

Name Start CMD Data

Length Index, Address, or Data Stop Check

Sum

Format HEX

GW MD 51 CMD Length A3 [1n-1]

GW MD 51 ACK Length A5 [1n-1]

ie. Check sum means the summation from byte 1 to byte n-1 with 8-bit length Baud rate 19200

Data bits 8

Parity No

Start bit 1

Stop bit 1 Medical Device /

Sensor

(MD) Gateway / Hub /

Computer

(GW) TX

RX

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2.2. Command List

Message name CMD/ACK Direction Length

(Bytes) Note

Read device clock time 0x23 GW MD 8

0x23 GW MD 8

Read device model 0x24 GW MD 8

0x24 GW MD 8

Read device serial number, part 1 0x27 GW MD 8

0x27 GW MD 8

Read device serial number, part 2 0x28 GW MD 8

0x28 GW MD 8

Read storage number of data 0x2B GW MD 8

0x2B GW MD 8

Write device clock time 0x 33 GW MD 8

0x33 GW MD 8
Turn off the device 0x50 GW MD 8
0x50 GW MD 8
Clear/Delete all memory 0x52 GW MD 8
0x52 GW MD 8

Notification for entering communication
mode 0x54 GW MD 8
Read data 0x71 GW MD 7
0x71 GW MD 34
Set user profile in the device 0x72 GW MD 12
0x72 GW MD 12

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2.2.1. [0x23] Read device clock time (4 bytes)
Message Name: Read device clock time (4 bytes)
Message ID: 0x23
Message Description: Read device date and time (to minute)
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Please refer to device clock time section for the format of date and time.

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read System Clock time GW MD 0x23 0x0 0x0 0x0 0x0
Response from device. GW MD 0x23 Day + Month + Year Minute Hour

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x23 GW commands MD to response device clock time
DATA 4 0x0 NA

Field Size
(Bytes) Value Description.
ACK 1 0x23 MD response device clock time
DATA 2 Data_1 + Data_0 : Day+Month+Year (word)
DATA 2 Data_2 : Minute, Data_3 : Hour

2.2.2. [0x24] Read device model
Message Name: Read device model
Message ID: 0x24
Message Description: Read device model / project code
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes
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Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read System Clock time GW MD 0x24 0x0 0x0 0x0 0x0
Response from device. GW MD 0x2 4 Project Code

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x24 GW commands MD to response device model
DATA 4 0x0

Field Size

(Bytes) Value Description.

ACK 1 0x 24 MD response device clock time
DATA 2 Data_1 + Data_0 : Device model in four digits (Data_1 is
the higher byte)
DATA 2 Data_3, Data_2 : undefined

2.2.3. [0x27] Read device serial number, part 1

Message Name: Read the device serial number, part 1

Message ID: 0x2 7

Message Description: Read the device serial number, part 1. This command cooperates
with 0x28 to complete the serial number.

Src / Des: GW / MD

Length: Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) :
SN_7 + SN_6 + SN_5 + SN_4 + SN_3 + SN_2 + SN_1 + SN_0 (example: 32 50 21 02 40 13 10 2A)

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3
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Read device serial

number, part 1 GW MD 0x27 0x0 0x0 0x0 0x0

Response from device GW MD 0x27 SN_0 SN_1 SN_2 SN_3

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x27 GW commands MD to response the device serial
number, part 1
DATA 4 0x0

Field Size

(Bytes) Value Description.

ACK 1 0x27 Read the device serial number , part 1
DATA 4 Data_0 ~ Data_3 : SN_0 ~ SN_3

2.2.4. [0x28] Read device serial number, part 2

Message Name: Read the device serial number, part 1

Message ID: 0x28

Message Description: Read the device serial number, part 2. This command cooperates with 0x27 to complete the serial number.

Src / Des: GW / MD

Length: Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) :

SN_7 + SN_6 + + SN_0 (example: 32 50 21 02 40 13 10 2A)

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Read device serial

number, part 2 GW MD 0x28 0x0 0x0 0x0 0x0

Response from device GW MD 0x28 SN_4 SN_5 SN_6 SN_7

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Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x28 GW commands MD to response the device serial number, part 2

DATA 4 0x0

Field Size

(Bytes) Value Description.

ACK 1 0x28 Read the device serial number, part 2

DATA 4 Data_0 ~ Data_3 : SN_4 ~ SN_7

2.2.5. [0x2B] Read storage number of data

Message Name: Read storage number of data

Message ID: 0x2B

Message Description: Read data counts

Src / Des: GW / MD

Length: Request: 8 bytes / Response: 8 bytes

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Read Storage Number

of Stored Readings GW MD 0x2B 0x0 0x0 0x0 0x0

Response from device GW MD 0x2B Storage Number 0x0 0x0

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x2B GW commands MD to response the Number of Stored Readings

DATA 4 0x0

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Field Size
(Bytes) Value Description.
ACK 1 0x2B MD response the the Number of Stored Readings
DATA 2 Data_1+Data_0 = Storage Number (Word)

2.2.6. [0x33] Write System Clock time (4 bytes)
Message Name: Write device clock time (4 bytes)
Message ID: 0x33
Message Description: Write the device date and time (t o minute)
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Please refer to device clock time section for the format of date and time.

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Write System Clock
time GW MD 0x33 Day+Month+Year Minute Hour
Response from device. GW MD 0x33 Day+Month+Year Minute Hour

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x33 GW commands MD to write System Clock time (4
bytes)
DATA 2 Data_1 + Data_0 : Day+Month+Year (word)
DATA 2 Data_2 : Minute, Data_3 : Hour

Field Size
(Bytes) Value Description.
ACK 1 0x33 MD write System Clock time (4 bytes) O.K.
DATA 2 Data_1 + Data_0 : Day+Month+Year (word)
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DATA 2 Data_2 : Minute, Data_3 : Hour

2.2.7. [0x50] Turn off the device
Message Name: Turn off the device
Message ID: 0x50
Message Description: Turn off the device from GW
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Turn off the device GW MD 0x50 0x0 0x0 0x0 0x0
GW MD 0x50 0x0 0x0 0x0 0x0

Field Description

Field Size
(Bytes) Value Description.
CMD 1 0x50 GW commands MD to turn off the power.
DATA 4 0x0 NA

Field Size
(Bytes) Value Description.
ACK 1 0x55 MD responses.
DATA 4 0x0 NA

2.2.8. [0x52] Clear/Delete all Memory
Message Name: Clear/Delete all memory
Message ID: 0x52
Message Description: Delete all measurement data in the device
Src / Des: GW / MD
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Length : Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Clear/Delete all Memory
Storages GW MD 0x52 0x0 0x0 0x0 0x0
Response from device. GW MD 0x52 0x0 0x0 0x0 0x0

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x52 GW commands MD to Clear/Delete all Memory
Storages in Device EEPROM
DATA 4 0x0 NA

Field Size
(Bytes) Value Description.
ACK 1 0x52 Clear/Delete all Memory Storages in Device EEP ROM
O.K.
DATA 4 0x0 NA

2.2.9. [0x54] Notification for entering communication mode
Message Name: Notification for entering communication mode
Message ID: 0x54
Message Description: Some models will send 0x54 command to host as a notification to indicate meter is ready for command from host. Host has to deal with this kind of notification in the beginning of connection (like RS-232) and prevent it to be mixed with the expect command response in receiving queue of host.
Src / Des: MD / GW
Length: Request: 8 bytes / Response: NA

Format
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Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Notification of entering
communication mode GW MD 0x54 0x0 0x0 0x0 0x0
No need to response from GW side

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x54 MD reports to GW that it enters communication
mode.

DATA 4 0x0 NA

2.2.10. [0x71] Read Data

Message Name: Read data

Message ID: 0x71

Message Description: Read measurement data

Src / Des: GW / MD

Length : Request: 7 bytes / Response: 34 bytes

Format

Byte 2 3 4 5 6 7

Message name Direction CMD Length Idx_L Idx_H Stop CheckSum

Read data GW MD 0x71 0x02 Low_index High_index A3 [1..6]

Byte 2 3 4~32 33 34

Message name Direction CMD Length Data Stop ChekSum

Response

from device GW MD 0x71 0x1D Hex A5 [133]

Field Description

Field Size

(Bytes) Value Description

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CMD 1 0x 71 Retrieve data in device with data index

DATA 2 HEX Idx_H + Idx_L = index of the data (word), start from
0. The value zero means newest record.

Field Size

(Bytes) Value Description

ACK 1 0x71 Response data from MD to GW

Length 1 0x1D Length of the data bytes

4 Stable time 1 This field is only for Test N'GO Scale 550.

Stable time. This unit is second and value range is
form 0 to 255.

5 Year 1 Year, the year in A.D. format

6 Month 1 Month

7 Day 1 Day

8 Hour 1 Hour, in 24-hour format
 9 Minute 1 Minute
 10 Code 1 Code / User ID, start from 1
 11 Gender 1 Gender. Female = 0, Male =1
 12 Height(cm) 1 Height in cm
 13 Height(in) 2 Height in inch x10. Byte 13 is the higher byte; Byte 14 is the lower byte
 15 Age 1 Age, integer value
 16 Cal unit 1 Calibration unit which means the current unit of scales. Kg=0, lb=1, st=2
 17 Weight(kg) 2 Weight in kg x10. Byte 17 is the higher byte
 19 Weight(lb) 2 Weight in lb x10. Byte 19 is the higher byte
 21 BMI 2 BMI % x10. Byte 21 is the higher byte
 23 BMR 2 Basal Metabolic Rate, in Kcal per day. Byte 23 is the higher byte
 25 Bf 2 Body Fat. Bf% x10. Byte 25 is the higher byte
 27 Bm 2 Body muscle. Bm% x10. Byte 27 is the higher byte
 29 Bn 1 Body bone. Bn% x10.
 30 Bw 2 Body water. Bw% x10. Byte 30 is the higher byte
 32 Status 1 Only valid for body fat model. Severely Underweight=1, Underweight=2, Normal=3, Foracare Inc.
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 Overweight =4, Obese =5

2.2.11. [0x72] Set user profile in the device
 Message Name: Set user profile in the device
 Message ID: 0x72
 Message Description: Set user profile from GW to MD
 Src / Des : GW / MD
 Length: Request: 12 bytes / Response: 12 bytes

Format
 Byte 2 3 4~10 11 12
 Message name Direction CMD Length Data Stop ChekSum
 Set user profile GW MD 0x 72 0x 07 Hex A3 [1 11]

Byte 2 3 4~10 11 12
 Message name Direction CMD Length Data Stop ChekSum
 GW MD 0x 72 0x 07 Hex A5 [1 11]

Field Description
 # Field Size
 (Bytes) Value Description
 2 CMD 1 0x72 Set user profile in the device.
 3 Length 1 0x07 Length of the data segment
 4 Code 1 Code / User ID, start from 1.
 5 Gender 1 Gender. Female = 0, Male =1
 6 Height(cm) 1 Height in cm. You have to input both in cm and in inch at the same time.
 7 Height(in) 2 Height in inch. Byte 7 is the higher byte; Byte 8 is the lower byte
 9 Age 1 Age , integer value

10 --- --- 1 No function

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#	Field	Size	(Bytes)	Value	Description
2	ACK	1	0x72		Response data from MD to GW
3	Length	1	0x07		Length of the data bytes
4	Code	1			Code / User ID, start from 1.
5	Gender	1			Gender. Female = 0, Male =1
6	Height(cm)	1			Height in cm
7	Height(in)	2			Height in inch. Byte 13 is the higher byte; Byte 14 is the lower byte
9	Age	1			Age, integer value
10	---	---	1	---	---

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Data Format

2.2.12. Device clock time (4 bytes)

Direction	CMD/	ACK	Data_0	Data_1	Data_2	Data_3	Description
GW	MD	0x23	0x0	0x0	0x0	0x0	Read device date and time
GW	MD	0x23	Day+Month+Year (ref:TableA) Minute+Hour (ref:TableB)				Response from device

Table A: Day + Month + Year

Table B: Minute and Hour

MSB	LSB														
Byte 1	Byte 0														
Year (7-bit) Month (4-bit) Day (5-bit)															
6	5	4	3	2	1	0	3	2	1	0	4	3	2	1	0
MSB	LSB														
Byte 3	Byte 2														
Hour (5-bit) Minute (6-bit)															
4	3	2	1	0	5	4	3	2	1	0					